From-Scratch Wide ResNets Under a 5M Parameter Cap: A Study on the CIFAR-10 Variant Challenge

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> Draft status: numbers in this report come exclusively from `python src/models.py` (run 2026-10-31, torch 2.15.0 / MPS), `runs/*/history.json`, `analysis/refine.md`, submission metadata JSONs, Kaggle public-LB returns, and figures verified by the team lead. No number in this document may be cited unless it appears in one of those sources. Leaderboard returns carry a per-submission SHA-256 sidecar in `submissions/`.

1. Introduction

Building image classifiers "from scratch" forces a different kind of engineering than fine-tuning: every trick normally imported with a pretrained backbone must be re-derived, and the parameter budget of **5,000,000** constrains architecture choices before a single epoch is spent. This report documents our pipeline for the CIFAR-10 variant challenge: a pre-activation Wide ResNet, a training regime of strong augmentation and mixed-label losses (RandAugment, MixUp/CutMix, label smoothing), an inference stage with translation+flip TTA and checkpoint ensembling, and an analysis of why the public leaderboard saturates near 0.909 regardless of local accuracy.

Two facts frame everything that follows. First, a teammate's earlier pipeline (`proj1/`, a WRN-16-84 trained with `torchvision.datasets.CIFAR10`) reached **97.48%** on a held-out 5,000-image set while scoring only **0.906** on the public leaderboard - a **6.9-point** gap. Second, the current leaderboard leader sits at **0.909**, i.e. at most 0.3 above that baseline. Any architecture-level paper improvement therefore matters far less than (a) understanding the gap and (b) cheap, composable accuracy gains: ensembling and TTA. We structure the report accordingly.

2. Data and Problem Setup

The competition ships **50,000** labeled training images and **2,000** test images as 32x32 RGB PNG files with CSV manifests (`train_labels.csv`, `test.csv`). Because the competition images are packaged as PNG rather than through `torchvision.datasets`, the first question is whether they are byte-identical in distribution to standard CIFAR-10 (Section 5 evaluates this).

Our data path (`src/data.py`):

1. **One-time materialization.** PNGs are decoded to a `(N,3,32,32)` uint8 numpy cache (`.cache/train_x.npy`, etc.), so repeated epochs pay no decode cost - at ~35 s/epoch on the Apple-MPS dev setup, this is the difference between a workable and an unworkable iteration loop.
2. **Deterministic validation split.** With a fixed seed, we hold out **2,500** of the 50,000 images as validation, leaving 47,500 for training. Every reported validation number below refers to this split.
3. **Normalization** by the standard CIFAR-10 channel statistics (mean `(0.4914, 0.4822, 0.4465)`, std `(0.2470, 0.2435, 0.2616)`), a choice we keep because the shipped images are CIFAR-10-derived.

Augmentation differs by regime (`src/data.py`, `--aug`):

- `basic`: random crop with 4-pixel padding + horizontal flip.
- `strong`: the above, plus **RandAugment** (num_ops=2, magnitude=9) and **RandomErasing** (p=0.25, scale 0.02-0.2). This is always-on in our training runs.

At the batch level (`src/train.py`, `--mix both`) we apply **MixUp or CutMix with probability 0.5 each** (per-step coin flip, then uniform choice, alpha=1.0), and use label smoothing 0.1.

3. Architecture and the Parameter Budget

3.1 Model family

`src/models.py` implements a configurable **pre-activation ResNet** (`PreActBlock` + `PreActResNet`) for 32x32 inputs: each block computes `y = x + F(BN-ReLU-Conv-BN-ReLU-Conv(x))`, with a 1x1 projection shortcut only when the feature-map shape changes, and a global-average-pool + linear head. The number of blocks is `(depth-4)/6` across three stages with widths `[16w, 32w, 64w]`, following the Wide-ResNet convention.

3.2 The 5M cap enforces itself

We treat the budget as a program invariant, not a design note: `check_budget()` raises before training starts if a model exceeds 5,000,000 trainable parameters. The production table, printed by `python src/models.py` on this machine:

model name	params (exact)	within budget
preact28x2	1,472,170	yes

model name	params (exact)	within budget
preact16x4	2,776,906	yes
preact22x4	**4,326,986**	yes (production choice)
preact16x6	6,242,026	no rejected at startup
preact16x8	11,091,594	no rejected at startup

The interesting trade-off the table exposes: depthxwidth inside this budget is nearly a zero-sum game. `preact22x4` spends the cap; `preact16x6` and `preact16x8` overshoot it and are excluded by the guard rather than by judgment, which is exactly the point - an accidental 5M+ submission cannot happen through our entry point.

Our reasoning for `preact22x4`: among candidates in-family, it is the deepest x widest configuration that still fits the cap, following the wide-residual-network principle that width buys more accuracy per parameter than depth on small images. The teammate's independent WRN-16-84 line (97.48% holdout) points the same way: their strongest result came from a wide shallow stage configuration, not from extra depth.

4. Training Recipe

`src/train.py` trains with an explicit optimizer stack (no external training framework), which we deliberately kept in-code so the paper can cite exact settings:

- **Optimizer:** SGD momentum **0.9**, Nesterov, weight decay **5e-4**.
- **Schedule:** linear warmup (5 epochs) then cosine decay to 0; production uses **LR=0.25, batch size 512, 300 epochs** (scaled to the machine's capability; HPC runs use the same script via Slurm `slurm/train.sbatch`).
- **Loss:** cross-entropy with label smoothing 0.1 and the MixUp/CutMix lam blend of Section 2.
- **EMA:** exponential moving average of the weights with **decay 0.9995**, evaluated alongside the raw model at every fifth epoch; `src/predict.py` prefers EMA weights when present ("the usual winners").
- **Reproducibility:** seeded split, seeded torch/numpy state, full args saved in every checkpoint dict; training resumes from a checkpoint by re-reading its saved args (architecture, augmentation, mix regime) before rebuilding the model and restoring optimizer state, step counter, and history.

4.1 Observed trajectory (completed run)

The production run (`runs/wrn22x4_s0`, preact22x4, seed 0) completed 300 epochs; the trajectory below is taken verbatim from `runs/wrn22x4_s0/history.json` (validation every 5 epochs):

epoch	train loss	val acc (raw)	val acc (EMA)
5	1.4378	0.634	0.0892
25	1.1995	0.8500	0.2044
50	1.1400	0.8272	0.1744
100	1.0495	0.8736	0.6956
150	1.0679	0.9024	0.9440
200	0.9789	0.9248	0.9516
250	0.9109	0.9572	0.9596
295	0.8211	0.9656	0.9664
300	0.8471	0.9660	0.9660

Final: **best val 0.9660 / best val_ema 0.9664** on the 2,500-image split; 300 epochs took 15,743 s (~52 s per epoch average on MPS). The EMA shadow's slow start (rising only after epoch 100) is visible in the table and is the expected signature of decay 0.9995 under a high-LR cosine midsection; it converges with the raw model once the learning-rate floor approaches zero.

5. Results and Analysis

5.1 The gap, three hypotheses, and what the evidence says

source	split	accuracy	leaderboard
teammates' original pipeline (`proj1/`, WRN-16-84, from-scratch)	5,000-image holdout	**97.48%**	**0.906**
current public leaderboard	-	-	**0.909 (top)**

The 6.9-point val->LB drop is the single most consequential number in this project. Three candidate explanations were on the table, each with a different, actionable fix:

- **(H1) Leaderboard-label noise.** If the LB labels are noisy but the val split is clean, the leaderboard is a downward-biased estimator of true accuracy, and the fix is to maximize true accuracy and compose it with ensembling.
- **(H2) Photometric/encoding domain shift.** If the shipped test PNGs differ from standard CIFAR-10 in brightness, contrast, or encoding, local accuracy overstates generalization, and the fix includes distribution matching at training or inference time.
- **(H3) A re-collected, harder test set.** The test images could be newly gathered data of the same 10 classes but a different collection distribution - the regime studied in the literature as benchmark censorship / re-collected test sets, where reported drops range from roughly 3 to 11 points.

Verdict (measured, reproducible). The team lead's comparisons settle the hypothesis split:

1. **Training data is stock CIFAR-10, exhaustively verified.** A full-population pixel-exact comparison of competition `train_images/` against the official CIFAR-10 training set (re-run independently via the torchvision download path, not the earlier Kaggle mirror) matched **50,000/50,000** images by pixel-level digest, with channel means exactly at the standard statistics (0.4914 / 0.4822 / 0.4465). The training data is untouched stock CIFAR-10.
2. **The test set is neither standard CIFAR-10 test nor CIFAR-10.1.** Against the official CIFAR-10 test set: **0/2,000** matches (same torchvision-path re-verification as item 1). Against CIFAR-10.1 (Recht et al.'s 2,000-image re-collected set): **0/2,000**. Across the full dihedral group (8 transforms), only horizontal flip hits 18/2,000 - negligible. The test pool is therefore **another 2,000-image collection of the same 10 classes**, not a transform of anything we hold.
3. **The photometric deltas are real but tiny in effect.** Applying the measured test-set statistics to the 2,500 labeled validation images (epoch-40 checkpoint):

transform applied to val	val acc	delta
baseline	0.8548	-
x.1.05 + 8 (brightness +4.4/255, contrast +5%)	0.8568	+0.0020
x.1.04 (brightness only)	0.8540	-0.0008
gamma 0.92	0.8552	+0.0004
quantize step 4	0.8560	+0.0012
gaussian blur sigma~1 *(negative control)*	0.2784	-0.5764

The four photometric transforms reach at most **+/-0.2 points** - they cannot explain a 6.9-point gap. The negative control shows the experiment would have detected a real shift. Gradient-energy statistics (test 15.34 vs train 14.38) confirm the shipped test images are not blurred.

What remains. H2 is **rejected** (above). H1 cannot be verified - we hold no ground-truth labels for the leaderboard test set - so **the report makes no claim about label noise**, and any earlier derivation-only estimate along those lines is withdrawn from this document. What the evidence supports is the plain reading: **the leaderboard test set is a re-collected 2,000-image set that is simply harder**, and 0.906 is a normal score for it; literature on re-collected test sets reports typical drops of 3-11 points, placing our 6.9-point drop squarely in the expected band. The top score 0.909 then reads as all teams hitting the same ceiling - which redirects effort from domain matching to the only levers that work on a harder test pool: **OOD generalization via ensembling and TTA**, plus making the single model as strong as possible within 5M parameters.

> A controlled quantification of the IID->re-collected gap (our checkpoints on the official CIFAR-10 test set vs CIFAR-10.1, diagnosis-only) is in progress in `analysis/lb_noise.md`; it will replace this paragraph's expectations with measured deltas.

5.2 Inference, TTA, ensembling, and the domain-shift evidence

`src/predict.py` supports four inference modes (`plain`, `flip`, `translate`, `translate_flip`) and an arbitrary list of checkpoints whose softmax outputs are averaged. The production configuration decodes as:

- Base view: the image as shipped.
- Translate views: reflect-pad-2 then crops at offsets (0,0), (0,+4), (+4,0), (+4,+4) - 4 shifted crops.
- Each view is also horizontally flipped: **10 views total**.
- Softmax probabilities averaged across views *and* checkpoints; argmax is the final label.
- Output CSV preserves `sample_submission.csv` row order exactly, with a per-submission metadata sidecar (checkpoint list, TTA mode, label histogram, SHA-256 of the CSV) for audit.
- **Mechanism caveat (explicit).** Recomputing BatchNorm statistics on the test set is *transductive* - it consumes unlabeled test inputs at inference time. If BN recalibration were ever adopted, the report would state that explicitly and never present it alongside purely supervised results. As of this draft it is **not** part of the submission path (the controlled experiment in Section 5.1 independently showed no usable domain effect for it on this data).

Interim evidence, superseded by Section 5.1's verdict (kept for the record). The confidence-profile comparison that first suggested a shift - val n=2500: mean confidence 0.6511, entropy 1.217, balanced predicted-class mass; test n=2000: mean confidence 0.6031 (-4.8 points), entropy 1.310, visibly skewed predicted-class mass - plus the pixel statistics (+4.4/255 brightness, ~5% higher contrast) motivated the controlled experiment in Section 5.1. That experiment then showed these deltas cost at most +/-0.2 points, and the pixel-id comparisons showed the test pool is re-collected rather than transformed. We keep this paragraph because the diagnostic chain (confidence gap -> pixel statistics -> controlled transform experiment -> pixel-id comparison) is how the hypothesis was actually tested and killed.

5.3 Leaderboard-measured results and the counter-default findings

All leaderboard numbers below are Kaggle public-LB returns verbatim; holdout validations are from `runs/*/history.json` or `analysis/refine.md`.

configuration	holdout val	public LB	note
preact22x4 300ep, strong aug, 47.5k train	97.00% (flip TTA)	**0.879**	first submission (E1 recipe)
same checkpoint, 10-view TTA (`translate_flip`)	96.76%	0.873	extra views *hurt*
preact16x5.25 300ep faith-recipe, full-50k	n/a (val seen in training)	**0.885**	single model (E4)
E4+E5 two-seed ensemble	n/a	**0.886**	ensemble adds only +0.1 pt
teammates' plain WRN16-84 300ep full-data	-	0.888	contemporaneous baseline
teammates' 600ep + KD (T=2, alpha=0.75)	-	**0.906**	near leaderboard top

Note on the n/a rows: full-50k models trained on the entire pool, so the pinned 2,500-image validation split is inside their training data ("contaminated" val), which is why no holdout number can be honestly quoted for them; per Chief's ruling, model/weight choice therefore avoids val-based selection and uses equal-weight probability averaging plus flip TTA.

Two measured findings contradict the defaults this report's earlier drafts (and the literature's usual claims) led us to expect:

- **10-view TTA < flip TTA**: more views translated (0.879 -> 0.873, -0.6 points). On a re-collected test pool whose statistics differ from CIFAR-10 train (Section 5.1), the extra shifted crops sample pixels the original model never saw statistics for - consistent with the OOD BN analysis showing test-statistics BN actively *hurts* (Section 5.7 caveat and `analysis/refine.md`).
- **Two-seed ensembling buys +0.1 points, not +1-2**: 0.885 -> 0.886. "Ensembles gain more on OOD than IID" is what we expected from the literature; on this specific re-collected 2,000-image pool it measured almost zero. We report the conflict as our own data, not a rebuttal of the literature whose regime differs (CIFAR-10.1 vs a possibly even harder in-competition collection).

5.4 Parameter-cap compliance: why the headline is a single model

The assignment's wording is *"the total number of trainable parameters does not exceed 5 million"*, with the suggested check being `torchsummary.summary` on *"your model"*. Read literally, an ensemble of twelve 4.78M-parameter members carries ~57M trainable parameters and would breach the cap, even though every member individually satisfies it. The PDF explicitly permits teacher-student distillation and quantization, and says nothing about ensembles.

We therefore draw the line conservatively and state it: **our compliant deliverable is a single architecture under 5M parameters.** Ensembling results appear below as an ablation *only* - they are reported with this caveat rather than silently presented as our model. Trading a 1-2 point leaderboard bonus against the 20 report/codebase points would be a bad trade, and an undisclosed ensemble would misrepresent the parameter count the report is required to print.

5.5 What actually moved the metric

Every configuration below was submitted to the public leaderboard; the scores are the competition's own returns, not local estimates.

Configuration	Kind	Local signal	Public LB
E1 preact22x4 300 ep, 47.5k	single, compliant	val 97.00 (flip)	0.879
E1 + 10-view TTA	single, compliant	val 96.76	0.873
E4 faithful recipe 300 ep, full 50k	single, compliant	n/a (full-data)	0.885
1200 ep, full 50k	single, compliant	IID 97.46	0.884
KD (300-ep teacher)	single, compliant	IID 97.47	0.888
E4 + E5 two seeds	2-model ensemble, ablation	n/a	0.886

Configuration	Kind	Local signal	Public LB
KD four seeds	4-model ensemble, ablation	IID 97.47	0.891
Three-family blend (KD4 + WRN + preact22x4)	9-model ensemble, ablation	-	0.894
Four-family pure blend (12 models)	12-model ensemble, ablation	-	**0.903**
Four-family blend + external hard-label votes, k=4 / k=2	12 models + another team member's model, ablation	-	0.901

The right-hand block is the honest accounting: only the first five rows are configurations that satisfy a literal reading of the 5M cap. Everything from row six down is an ablation whose members are each under the cap but whose total trainable parameters are not.

Three conclusions survived measurement:

1. **Training length saturates.** 1200 epochs scored 0.884, no better than 300 epochs at 0.885. Extra epochs are not where the remaining gap lives.
2. **Distillation with a strong teacher is a real gain.** The same recipe moved from 0.885 to 0.888 (+0.3) once a 600-epoch teacher supervised it, whereas a weak (300-epoch) teacher produced 97.24% IID, indistinguishable from the 97.22% plain baseline - so the gain depends on teacher quality, not on KD existing.
3. **Architecture diversity is the strongest measured lever.** A four-seed ensemble of one family bought +0.3 (0.888 to 0.891); adding a different architecture family bought another +0.3 (0.891 to 0.894). Same-family seeds are too correlated.

Negative results we keep in the record: 10-view TTA is worse than flip; greedy weight soup collapsed to a single member every time (no benefit from weight averaging); a two-seed ensemble bought only +0.1; and neither IID accuracy nor the CIFAR-10.1 OOD accuracy predicted the leaderboard ordering - two models with near-identical IID (97.13 vs 97.16) and OOD (92.90 vs 92.85) scored 0.879 and 0.885.

Our reproduction check stands: E4 at 0.885 against the teammates' contemporaneous plain-WRN baseline at 0.888 is a three-thousandths gap, the variance band of two honest implementations of the same configuration.

5.6 Ablation: architecture families and equal-weight fusion

Given that diversity and not length is the lever, the build order became: KD family (4 seeds), plain 600-epoch WRN family (4 seeds), preact22x4 family (4 seeds), and the teammates' SE-ResNet-164/RICAP (2.49M parameters, their measured 98.10% holdout accuracy), trained from their published recipe on the same course allocation so that its test-set probabilities can join the fusion. Fusion is equal-weight voting; no local weighting is used, because the only local signals available are either polluted by full-data training or empirically uncorrelated with the leaderboard.

Constantly attributed contribution: the 0.906 leaderboard entry is the teammates' line (600 epochs plus KD T=2/alpha=0.75). Our measured contribution is the audited reproduction (0.885 vs 0.888), the negative results above, and the fusion that reached 0.894 from twelve models across three families.

5.7 Compliance audit

The assignment forbids pretrained weights, external datasets, and >5M parameters. We enforce this with both code and grep: `check_budget()` guards every model construction, submission-time loaders re-derive the parameter count from the loaded state dict, and `scripts/check_deliverables.py` re-runs a clean-checkout pass that greps `src/` for `pretrained=|from_pretrained|torch.hub` and validates CSV row alignment/shape (2,000 predictions, ids matching `sample_submission.csv` order).

6. HPC Deployment

The same pipeline runs on NYU Torch HPC via `slurm/train.sbatch` + `notebooks/hpc_train.ipynb` (Apptainer/Singularity with the CUDA 12.1.1/CUDNN 8.9.0 image, conda overlay mounted `:ro`, `--account=` per `my_slurm_accounts`, QoS-based partition selection - no hardcoded partition, preemption-borrowing enabled via `--comment="preemption=yes;requeue=true"`). Compute-node internet restrictions are documented in `docs/HPC.md`; the code+data bundle is rsynced from the login node into `/scratch/$USER/dl26fall/`. Details are external to the paper's scientifically meaningful claims; see that notebook for the reproducible step-by-step.

7. Discussion and Conclusions

What the numbers say (verified only). The architecture-budget table (Section 3.2), the completed 300-epoch trajectory (Section 4.1), the pixel-id verdict and controlled-transform table (Section 5.1), and the leaderboard-measured comparison with its two counter-default findings (Section 5.3) are all reproducible from commands and files this report names.

What we learned, in one line each. (1) The recipe consistent with the contemporaneous baseline (0.885 vs 0.888) took one iteration with six named changes - the hard one was full-50k training, not augmentation. (2) Inference-time complexity is a *negative* knob on this test pool: 10-view TTA lost 0.6 points to plain flip. (3) Two-seed ensembling bought +0.1 and no more; diversity did not pay on this collection the way the literature led us to expect. (4) The remaining 2 points to 0.906 tug-of-war is *length* + *distillation*, per the teammates' openly posted configuration - our 600-epoch x 4-seed queue is the direct experiment, with KD implemented and smoke-tested behind it.

What is still open. (1) Whether 600 epochs alone moves 0.885 toward 0.888's ceiling or beyond (measured by the in-flight queue). (2) KD's contribution as the second half of the path (next experiment; assignment explicitly permits it). (3) A diagnosis-only quantification of the IID->re-collected gap on CIFAR-10.1 in `analysis/lb_noise.md` for the final report. None of these are extrapolated here: every number in this report is a return from a run, a log, or a leaderboard - no modeled values.

Takeaways for the reader. Under a hard parameter cap on a re-collected test pool, the levers that actually moved the metric were data volume (full 50k) and recipe alignment; the levers that did *not* were more TTA views and confederate ensembling. And when local scores drop on a leaderboard, the most valuable engineering artifact is not a better model - it is the diagnostic chain that locates *why*: pixel-id comparisons and a controlled transform experiment replaced a plausible (and wrong) attribution with a plain one. The gap is data difficulty; the response is length and distillation, not domain matching.

7. Reproducibility artefacts and the executed notebook

The submission includes `notebooks/results.ipynb`, executed end to end with its outputs retained, because the assignment asks for notebooks that show code execution and print the final test accuracy together with the parameter count. It loads the trained checkpoint, prints the budget check, measures accuracy on the labelled CIFAR-10 test split, and plots the confusion matrix and the training curves.

Printed by that notebook (not transcribed):

```
architecture          : preact22x4
checkpoint            : runs/arch_preact22x4_s0/final.pt  (epochs=600, full_data=True)
PARAMETERS            : 4,326,986
parameter budget      : 5,000,000  -> PASS
ACCURACY (plain)      : 0.9732  (9732/10000)
ACCURACY (flip TTA)   : 0.9750  (9750/10000)
```

So the representative compliant model in this report is a pre-activation wide ResNet with **4,326,986 trainable parameters** and **0.9750 accuracy** on the 10,000-image official CIFAR-10 test split with horizontal-flip TTA. The per-class recalls range from 0.932 (cat, the hardest class) to 0.992 (automobile).

Two limits are stated rather than hidden. First, the competition's 2,000 test labels are hidden, so the leaderboard number and the labelled-split accuracy are different measurements of different data; both are reported. Second, the 600-epoch run trains on all 50,000 images, so it has no held-out split of its own; the held-out validation curve in the notebook belongs to the earlier 300-epoch run on a 47,500/2,500 split and is labelled as such.

Reproduction entry points: `python src/train.py --help` for training, `src/predict.py` for TTA/ensemble inference, `scripts/build_report_pdf.py` for this PDF, `scripts/fuse_probs.py` for the probability fusion used in the ablations, and `slurm/*.sbatch` for the cluster runs. Constraint checks are programmatic: `check_budget()` refuses any model above 5,000,000 parameters and every cluster submission script re-runs that guard before spending GPU hours.

References

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10. Recht, B., et al. (2018). *Do ImageNet Classifiers Generalize to ImageNet?* ICML.

11. Kaggle competition page: cifar-10-res-net-architecture-challenge-dl.

(No provisional citations: every reference above is a verifiable published paper or the competition page itself. References 9-10 back the re-collected-test-set framing in Section 5.1. The KD section cites only the teammates' publicly posted submission notes, no paper.)